

Course- B.Tech.
Course Code- CSET343
Year- 4TH Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Lab Assignment - 7

Objective: The objective of this assignment is to use the Grid-Search Capability of Scikit-learn on a deep learning model.

Dataset: [Pima Indians Diabetes](#)

Task1: Data Acquisition and Loading

Task2: Data Preprocessing

(Note: Task1 and Task2 should be followed as in previous labs)

Task3: Deep Learning Grid Search

- Using Keras models in Scikit-learn design a deep learning model.
- Using Grid search in scikit-learn:
 - tune batch size and training epochs.
 - tune learning rate
 - tune network weight initialization.
 - tune activation functions.
 - tune dropout regularization.
 - tune the number of neurons in the hidden layer.